

Austin Okray

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EDUCATION

State University of New York at Buffalo

Master of Science in Data Science

Buffalo, NY

Aug. 2020 – Dec. 2021

University of Wyoming

Bachelor of Science in Computer Science, Minors in Math and Statistics

Laramie, WY

Aug. 2014 – May 2020

WORK EXPERIENCE

CACI International Inc. Shared Artificial Intelligence Laboratory

Denver, CO

Data Scientist

January 2022 – Present

- Designing and implementing extensible Python code and data structures to support team-wide capabilities
- Employing modern machine learning libraries such as PyTorch, Torchvision, RL4CO, and Transformers to achieve research objectives independently and in teams
- Constructing custom neural networks for unique problem settings and applications
- Engaging in applied research projects in machine learning and statistics, including:
 - * Quickly finding solutions to combinatorial optimization problems with reinforcement learning
 - * Developing and using statistical significance tests for determining the best model(s) from a set of models
 - * Detecting out-of-distribution instances (synthetic and natural) in CV and NLP domains
 - * Researching and writing a whitepaper evaluating the landscape/risks of adversarial AI
 - * Reducing bias in CV via sampling strategies, bespoke loss functions, and dataset augmentation

Markforged (formerly Teton Simulation Software)

Laramie, WY

Machine Learning/Optimization and Software Development Intern

October 2019 – August 2020

- Utilizing and developing machine learning and optimization methods for optimizing aspects of 3D printed parts
- Maintaining and developing software in Python critical to the infrastructure of the optimization

University of Wyoming Computer Science Department

Laramie, WY

Undergraduate Research Assistant

May 2018 – Jan. 2020

- Researching social bias reduction in machine learning models, topics include: feature selection and kernel methods for bias reduction
- Attending conferences, presenting original work, other academic papers, progress, and newly researched algorithms
- Implementing algorithms from academic papers and textbooks

University of Wyoming Information Technology

Laramie, WY

Database Developer

May 2017 – Sep. 2019

- Utilizing SQL, Groovy, Java, Javascript, HTML, REST APIs, and PL/SQL to implement and maintain projects
- Maintaining healthy code structure with Git, from the organization level to small team settings
- Cooperating frequently with clients to determine and fix problems across multiple departments

PUBLICATIONS

- A. Okray, H. Hu, C. Lan. Fair Kernel Regression via Fair Feature Embedding in Kernel Space, International Conference on Tools in Artificial Intelligence, 2019.
- Z. Wang, S. Muknahallipatna, M. Fan, A. Okray, C. Lan. Music Classification using an Improved CRNN with Multi-Directional Spatial Dependencies in Both Time and Frequency Dimensions, International Joint Conference on Neural Networks 2019.

PROJECTS

Fairness for All | Python, Flask, Javascript, PostgreSQL

Sep. 2019 – April 2024

- Allows users to learn about fair machine learning interactively, storing past results in a database for fast lookup

TECHNICAL SKILLS AND CERTIFICATES

Languages (in order of experience): Python, SQL, R, C/C++, JavaScript, HTML/CSS

Libraries: NumPy, PyTorch, Torchvision, Matplotlib, Seaborn, Pandas, scikit-learn, RL4CO, Transformers

Security Clearance: TS/SCI